

1. Make sure you can read the file

```
#include "library.h"
```

```
void read_file(istream & fin)
```

```
{
```

```
    int year, month, day;
```

```
    double min_temp, avg_temp, max_temp, snow, rain, wind;
```

```
    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain >> wind;
```

```
    if (fin.fail())
```

```
        return;
```

```
    cout << "The average temperature was " << avg_temp << " F" << endl;
```

```
    read_file(fin);
```

```
}
```

```
void main()
```

```
{
```

```
    ifstream fin("MIAMI-FL.txt");
```

```
    if (fin.fail())
```

```
    {
```

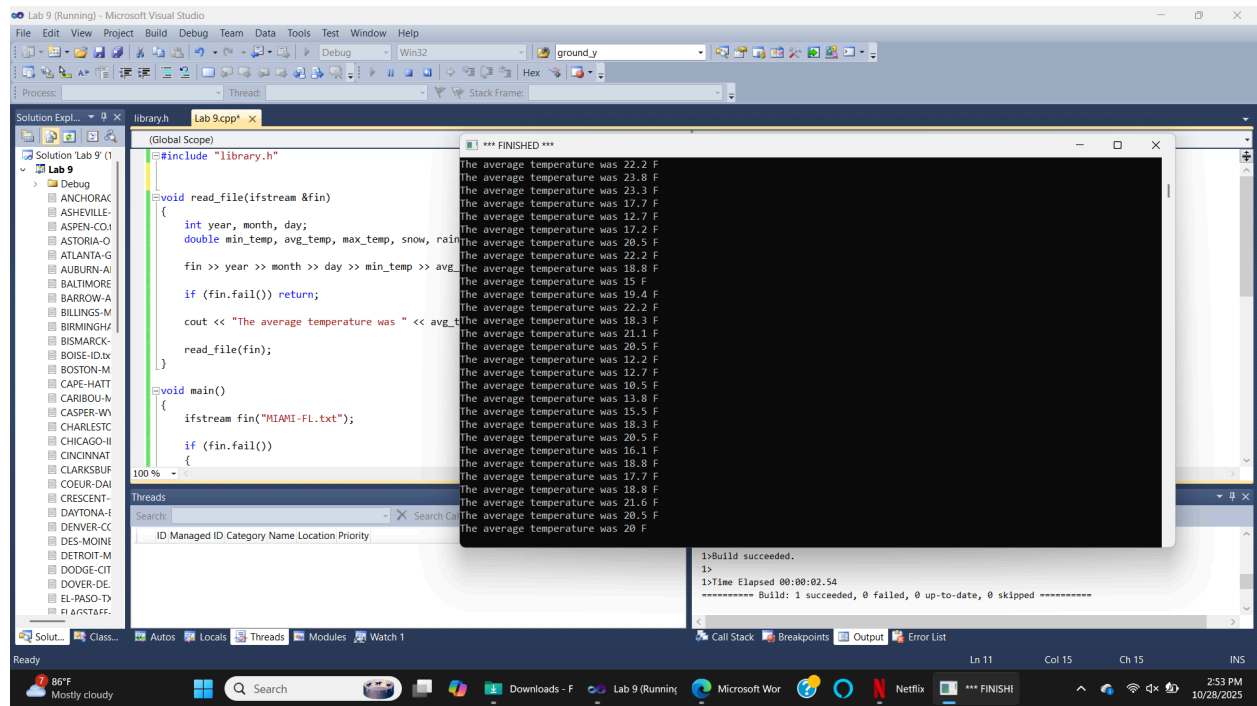
```
        cout << "The file didn't open!\n";
```

```
    }
```

```
    read_file(fin);
```

```
    fin.close();
```

```
}
```



2. Turn the numbers into a graph

```
#include "library.h"
```

```
void points(ifstream & fin)
{
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain >>
    wind;

    if (fin.fail())
        return;

    double x = (month * 31 + day) * 1.8;
    double y = 220 - avg_temp * 4;
    set_pen_width(3);
    set_pen_color(color::dark_green);
    draw_point(x,y);

    points(fin);
}
```

```
void graph()
{
    make_window(800,600);
    move_to(0,100);
    set_pen_color(color::orange);
    turn_right_by_degrees(90);
    draw_distance(900);
    note_position();
    move_to(0,185);
    draw_distance(900);
}
```

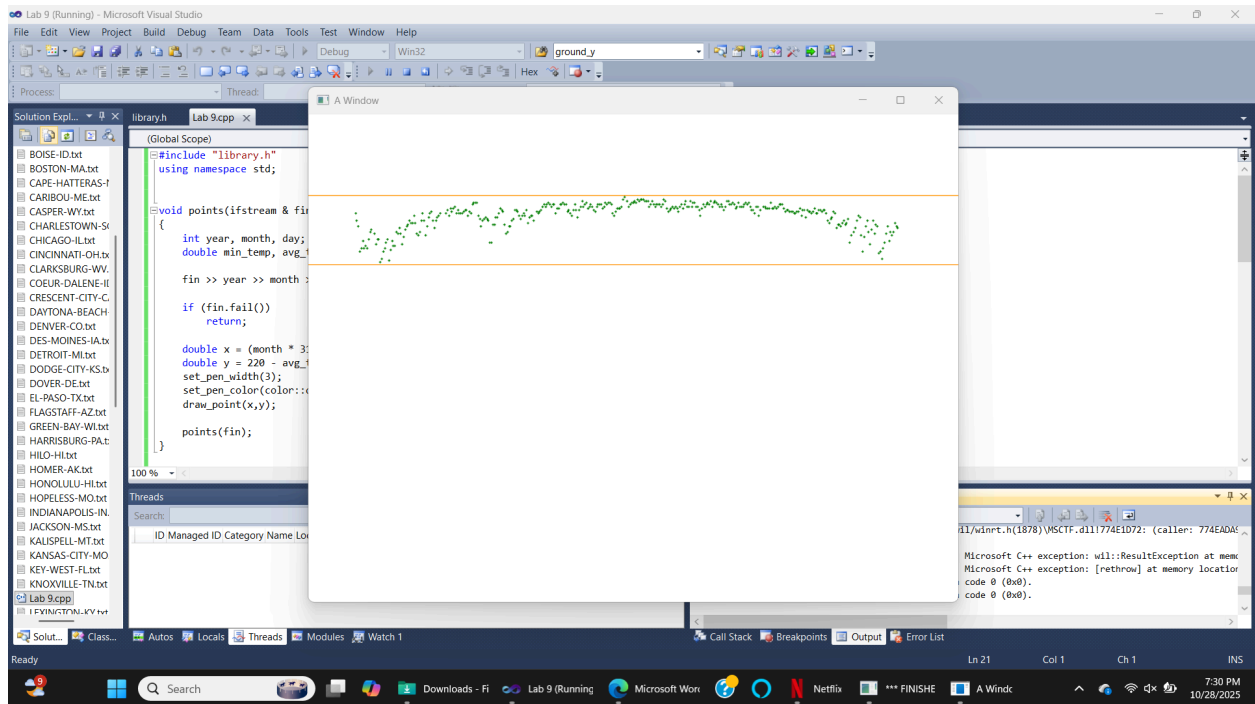
```
void main()
{
    graph();
    ifstream fin("MIAMI-FL.txt");
```

```

if (fin.fail())
{
    cout << "The file didn't open!\n";
}

points(fin);
fin.close();
}

```



3. Get the X coordinates right

```
#include "library.h"
```

```
int length_of_a_month(const int month)
```

```
{
    if (month == 1)
        return 31;
    if (month == 2)
        return 28;
    if (month == 3)
        return 31;
    if (month == 4)
        return 30;
    if (month == 5)
        return 31;
    if (month == 6)
        return 30;
    if (month == 7)
        return 31;
    if (month == 8)
        return 31;
    if (month == 9)
        return 30;
    if (month == 10)
        return 31;
    if (month == 11)
        return 30;
    if (month == 12)
        return 31;
}
```

```
int day_of_the_year(const int month, const int day)
```

```
{
    if (month == 1) return day;
    return length_of_a_month(month - 1) + day_of_the_year(month - 1, day);
}
```

```
void points(ifstream & fin)
```

```
{
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;
```

```
    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain >>
wind;
```

```
    if (fin.fail())
        return;
```

```
    double x = ((day_of_the_year(month,day)) * 1.8) + 50;
    double y = 220 - avg_temp * 4;
        set_pen_width(3);
        set_pen_color(color::dark_green);
        draw_point(x,y);
```

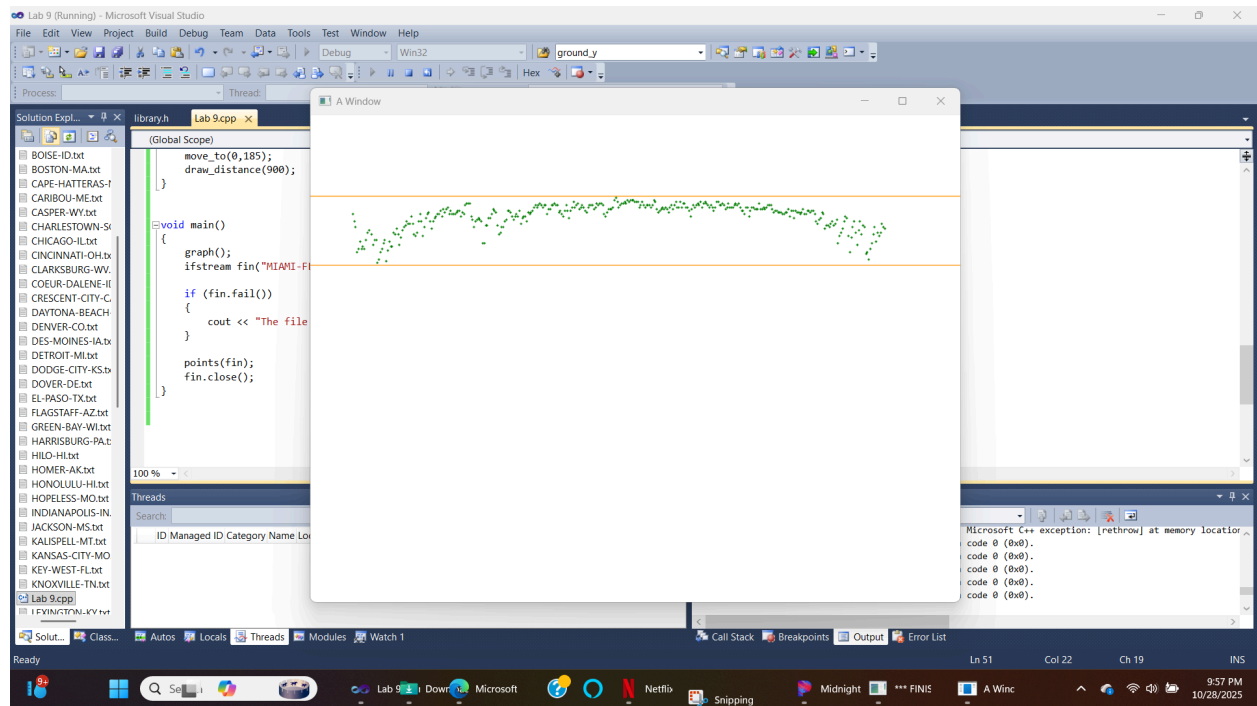
```
    points(fin);
}
```

```
void graph()
{
    make_window(800,600);
    move_to(0,100);
    set_pen_color(color::orange);
    turn_right_by_degrees(90);
    draw_distance(900);
    note_position();
    move_to(0,185);
    draw_distance(900);
}
```

```
void main()
{
    graph();
    ifstream fin("MIAMI-FL.txt");

    if (fin.fail())
    {
        cout << "The file didn't open!\n";
    }

    points(fin);
    fin.close();
}
```



4. Less dottiness

```
#include "library.h"
```

```
int length_of_a_month(const int month)
```

```
{  
    if (month == 1)  
        return 31;  
    if (month == 2)  
        return 28;  
    if (month == 3)  
        return 31;  
    if (month == 4)  
        return 30;  
    if (month == 5)  
        return 31;  
    if (month == 6)  
        return 30;  
    if (month == 7)  
        return 31;  
    if (month == 8)  
        return 31;  
    if (month == 9)  
        return 30;  
    if (month == 10)  
        return 31;  
    if (month == 11)  
        return 30;  
    if (month == 12)  
        return 31;  
}
```

```
    return 0;  
}
```

```
int day_of_the_year(const int month, const int day)
```

```
{  
    if (month == 1) return day;  
    return length_of_a_month(month - 1) + day_of_the_year(month - 1, day);  
}
```

```
void line(ifstream & fin)
```

```
{  
    int year, month, day;
```

```

double min_temp, avg_temp, max_temp, snow, rain, wind;

fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain >>
wind;

if (fin.fail())
    return;

double x = ((day_of_the_year(month,day)) * 1.8) + 50;
double y = 220 - avg_temp * 4;
    set_pen_width(3);
    set_pen_color(color::dark_green);
    draw_to(x,y);

line(fin);
}

void graph()
{
    make_window(800,600);
    move_to(0,100);
    set_pen_color(color::orange);
    turn_right_by_degrees(90);
    draw_distance(900);
    note_position();
    move_to(0,185);
    draw_distance(900);
}

void main()
{
    graph();
    ifstream fin("MIAMI-FL.txt");

    if (fin.fail())
    {
        cout << "The file didn't open!\n";
    }

    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

```

```

fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
>> wind;

```

```

double first_x = (day_of_the_year(month, day) * 1.8) + 50;

```

```

double first_y = 220 - avg_temp * 4;

```

```

move_to(first_x, first_y);

```

```

line(fin);

```

```

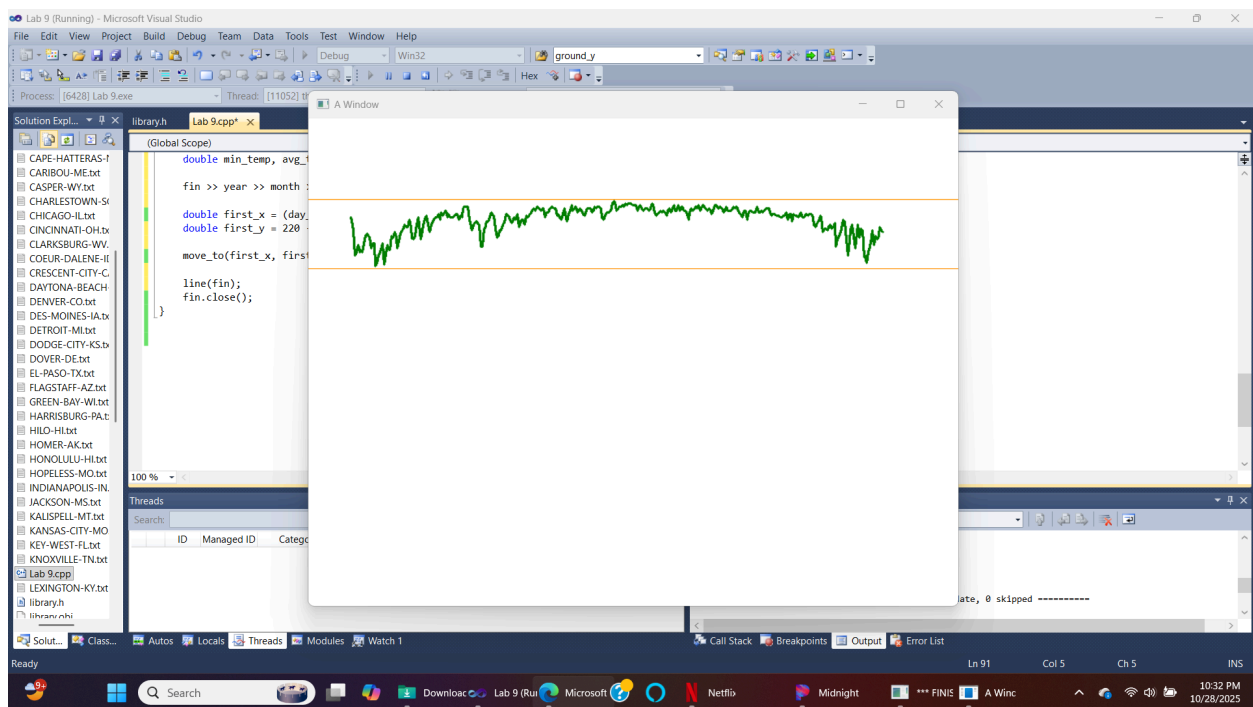
fin.close();

```

```

}

```



5. More information

```
#include "library.h"
```

```
int length_of_a_month(const int month)
{
    if (month == 1)
        return 31;
    if (month == 2)
        return 28;
    if (month == 3)
        return 31;
    if (month == 4)
        return 30;
    if (month == 5)
        return 31;
    if (month == 6)
        return 30;
    if (month == 7)
        return 31;
    if (month == 8)
        return 31;
    if (month == 9)
        return 30;
    if (month == 10)
        return 31;
    if (month == 11)
        return 30;
    if (month == 12)
        return 31;

    return 0;
}
```

```
int day_of_the_year(const int month, const int day)
{
    if (month == 1) return day;
    return length_of_a_month(month - 1) + day_of_the_year(month - 1, day);
}
```

```
void Temps(ifstream & fin, double Xmax, double Ymax, double Xmin, double Ymin)
```

```

{
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
    >> wind;

    if (fin.fail())
        return;

    double x = ((day_of_the_year(month,day)) * 1.8) + 50;
    double y_max = 240 - max_temp * 4;
    double y_min = 240 - min_temp * 4;

    set_pen_width(3);

    set_pen_color(color::red);
    move_to(Xmax, Ymax);
    draw_to(x, y_max);

    set_pen_color(color::blue);
    move_to(Xmin, Ymin);
    draw_to(x, y_min);

    Temps(fin, x, y_max, x, y_min);
}

```

```

void graph()
{
    make_window(800,600);
    move_to(0,100);
    set_pen_color(color::orange);
    turn_right_by_degrees(90);
    draw_distance(900);
    note_position();
    move_to(0,235);
    draw_distance(900);
}

```

```

void main()
{
    graph();
}

```

```

ifstream fin("MIAMI-FL.txt");

if (fin.fail())
{
    cout << "The file didn't open!\n";
}

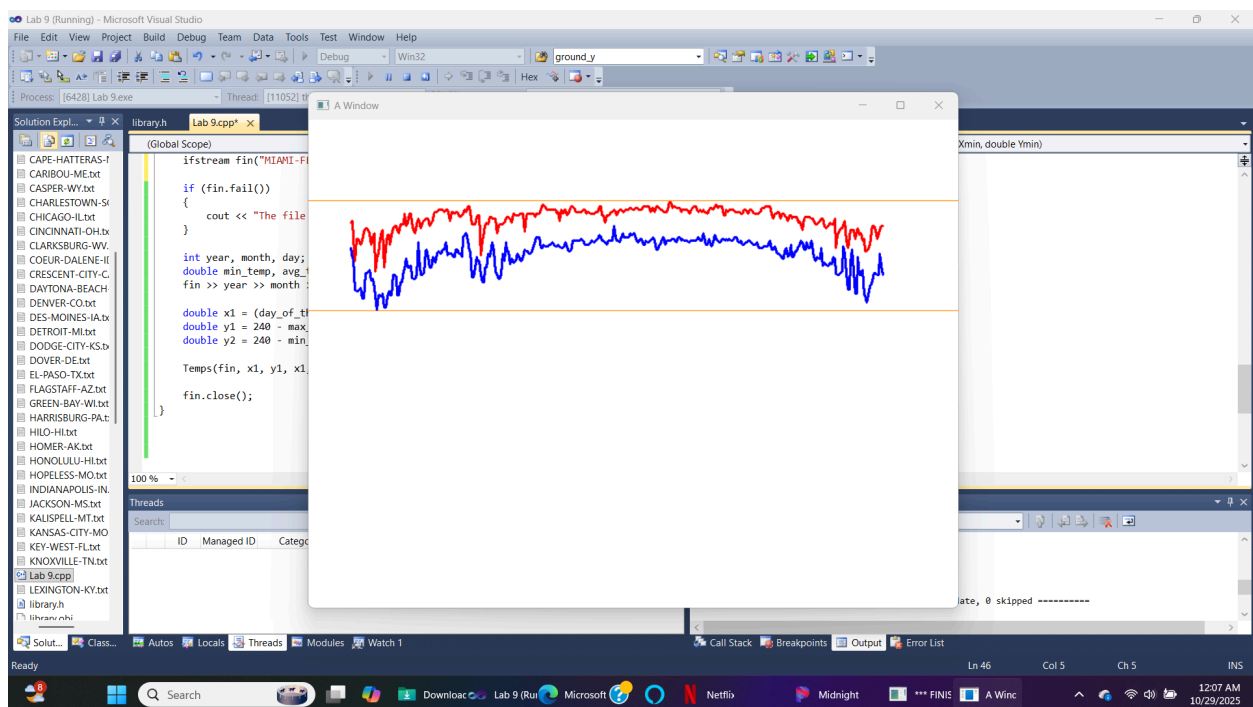
int year, month, day;
double min_temp, avg_temp, max_temp, snow, rain, wind;
fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
>> wind;

double x1 = (day_of_the_year(month, day) * 1.8) + 50;
double y1 = 240 - max_temp * 4;
double y2 = 240 - min_temp * 4;

Temps(fin, x1, y1, x1, y2);

fin.close();
}

```



6. Kites

```
#include "library.h"
```

```
int length_of_a_month(const int month)
```

```
{  
    if (month == 1)  
        return 31;  
    if (month == 2)  
        return 28;  
    if (month == 3)  
        return 31;  
    if (month == 4)  
        return 30;  
    if (month == 5)  
        return 31;  
    if (month == 6)  
        return 30;  
    if (month == 7)  
        return 31;  
    if (month == 8)  
        return 31;  
    if (month == 9)  
        return 30;  
    if (month == 10)  
        return 31;  
    if (month == 11)  
        return 30;  
    if (month == 12)  
        return 31;  
  
    return 0;  
}
```

```
int day_of_the_year(const int month, const int day)
```

```
{  
    if (month == 1) return day;  
    return length_of_a_month(month - 1) + day_of_the_year(month - 1, day);  
}
```

```

void Temps(ifstream & fin, double Xmax, double Ymax, double Xmin, double Ymin, double
Xwind, double Ywind)
{
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
>> wind;

    if (fin.fail())
        return;

    double x = ((day_of_the_year(month,day)) * 1.8) + 50;
    double y_max = 240 - max_temp * 4;
    double y_min = 240 - min_temp * 4;
    double y_wind = 400 - wind * 4;

    set_pen_width(3);

    set_pen_color(color::red);
    move_to(Xmax, Ymax);
    draw_to(x, y_max);

    set_pen_color(color::blue);
    move_to(Xmin, Ymin);
    draw_to(x, y_min);

    set_pen_color(color::light_blue);
    move_to(Xwind, Ywind);
    draw_to(x, y_wind);

    Temps(fin, x, y_max, x, y_min, x, y_wind);
}

```

```

void graph()
{
    make_window(800,600);
    move_to(0,100);
    set_pen_color(color::orange);
    turn_right_by_degrees(90);
    draw_distance(900);
    note_position();
    move_to(0,235);
}

```

```

        draw_distance(900);
    }

void main()
{
    graph();

    ifstream fin("MIAMI-FL.txt");

    if (fin.fail())
    {
        cout << "The file didn't open!\n";
    }

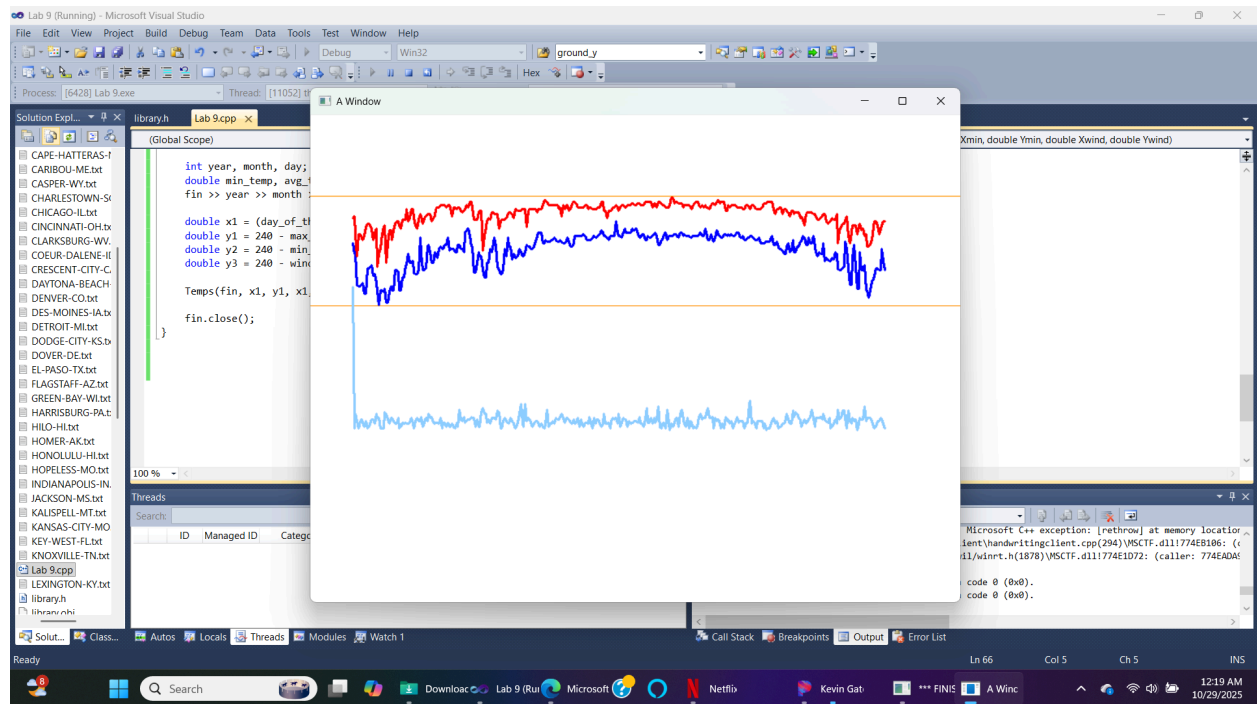
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;
    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
>> wind;

    double x1 = (day_of_the_year(month, day) * 1.8) + 50;
    double y1 = 240 - max_temp * 4;
    double y2 = 240 - min_temp * 4;
    double y3 = 240 - wind * 4;

    Temps(fin, x1, y1, x1, y2, x1, y3);

    fin.close();
}

```



7. Make it useful

```
#include "library.h"
```

```
int length_of_a_month(const int month)
```

```
{  
    if (month == 1)  
        return 31;  
    if (month == 2)  
        return 28;  
    if (month == 3)  
        return 31;  
    if (month == 4)  
        return 30;  
    if (month == 5)  
        return 31;  
    if (month == 6)  
        return 30;  
    if (month == 7)  
        return 31;  
    if (month == 8)  
        return 31;  
    if (month == 9)  
        return 30;  
    if (month == 10)  
        return 31;  
    if (month == 11)  
        return 30;  
    if (month == 12)  
        return 31;  
  
    return 0;  
}
```

```
int day_of_the_year(const int month, const int day)
```

```
{  
    if (month == 1) return day;  
    return length_of_a_month(month - 1) + day_of_the_year(month - 1, day);  
}
```

```

void Temps(ifstream & fin, double Xmax, double Ymax, double Xmin, double Ymin, double
Xwind, double Ywind, double & highest_temp, double & lowest_temp)
{
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
>> wind;

    if (fin.fail())
        return;

    if (max_temp > highest_temp)
        highest_temp = max_temp;
    if (min_temp < lowest_temp)
        lowest_temp = min_temp;

    double x = ((day_of_the_year(month,day)) * 1.8) + 50;
    double y_max = 240 - max_temp * 4;
    double y_min = 240 - min_temp * 4;
    double y_wind = 400 - wind * 4;

    set_pen_width(3);

    set_pen_color(color::red);
    move_to(Xmax, Ymax);
    draw_to(x, y_max);

    set_pen_color(color::blue);
    move_to(Xmin, Ymin);
    draw_to(x, y_min);

    set_pen_color(color::light_blue);
    move_to(Xwind, Ywind);
    draw_to(x, y_wind);

    Temps(fin, x, y_max, x, y_min, x, y_wind, highest_temp, lowest_temp);
}

```

```

void x_axis_month(int month)
{
    if (month > 12)
        return;
}

```

```
double x = (day_of_the_year(month, 1) * 1.8) + 50;
```

```
set_pen_color(color::grey);
```

```
move_to(x, 40.0);
```

```
draw_to(x, 440.0);
```

```
string initial;
```

```
if (month == 1)
```

```
    initial = "J";
```

```
if (month == 2)
```

```
    initial = "F";
```

```
if (month == 3)
```

```
    initial = "M";
```

```
if (month == 4)
```

```
    initial = "A";
```

```
if (month == 5)
```

```
    initial = "M";
```

```
if (month == 6)
```

```
    initial = "J";
```

```
if (month == 7)
```

```
    initial = "J";
```

```
if (month == 8)
```

```
    initial = "A";
```

```
if (month == 9)
```

```
    initial = "S";
```

```
if (month == 10)
```

```
    initial = "O";
```

```
if (month == 11)
```

```
    initial = "N";
```

```
if (month == 12)
```

```
    initial = "D";
```

```
double next_x = (day_of_the_year(month + 1, 1) * 1.8) + 50;
```

```
if (month == 12)
```

```
    next_x = (365 * 1.8) + 50;
```

```
else
```

```
    next_x = (day_of_the_year(month + 1, 1) * 1.8) + 50;
```

```
double letter = (x + next_x) / 2;
```

```
set_pen_color(color::magenta);
```

```

        set_font_size(20);
        move_to(letter - 5, 460.0);

        write_string(initial);
        x_axis_month(month + 1);
    }

```

```

void y_axis_temp(int temp)
{
    if (temp > 100)
        return;

    double y = 440 - temp * 4;

    set_pen_color(color::black);
    move_to(50.0, y);
    draw_to(709.0, y);

    set_pen_color(color::magenta);
    set_font_size(20);
    move_to(20.0, y);
    write_string(temp);

    y_axis_temp(temp + 10);
}

```

```

void max_and_min(double highest_temp, double lowest_temp)
{
    double y_high = 240 - highest_temp * 4;
    double y_low = 240 - lowest_temp * 4;

    set_pen_width(2);
    set_pen_color(color::orange);

    move_to(50.0, y_high);
    draw_to(709.0, y_high);
    move_to(50.0, y_low);
    draw_to(709.0, y_low);
}

```

```

void graph()

```

```

{
    make_window(800, 600);

    set_pen_color(color::black);
    move_to(50.0, 40.0);
    draw_to(50.0, 440.0);
    move_to(709.0, 40.0);
    draw_to(709.0, 440.0);

    x_axis_month(1);
    y_axis_temp(0);
}

```

```

void main()
{
    graph();

    ifstream fin("MIAMI-FL.txt");

    if (fin.fail())
    {
        cout << "The file didn't open!\n";
    }

    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
    >> wind;

    double x1 = (day_of_the_year(month, day) * 1.8) + 50;
    double y1 = 240 - max_temp * 4;
    double y2 = 240 - min_temp * 4;
    double y3 = 240 - wind * 4;
    double highest_temp = max_temp;
    double lowest_temp = min_temp;

    Temps(fin, x1, y1, x1, y2, x1, y3, highest_temp, lowest_temp);

    fin.close();

    max_and_min(highest_temp, lowest_temp);
}

```

The screenshot displays the Microsoft Visual Studio IDE with a C++ project named 'Lab 9 (Running)'. The main window shows the 'A Window' output, which is a line graph titled 'Temps(fin, x1, y1, x1, fin.close());'. The graph plots temperature data for 12 months (J, F, M, A, M, J, J, A, S, O, N, D). The y-axis ranges from 0 to 100. Three data series are shown: 'highest_temp' (red line), 'avg' (blue line), and 'lowest_temp' (light blue line). The 'highest_temp' series fluctuates between approximately 60 and 80, 'avg' between 50 and 70, and 'lowest_temp' between 10 and 20. The 'Solution Explorer' on the left shows the project structure, including 'Lab 9.cpp' and 'library.h'. The 'Output' window at the bottom right shows a Microsoft C++ exception message: 'Microsoft C++ exception: [rethrow] at memory location 0x00000000. (11774E1D72): (caller: 774EAD0A)'. The status bar at the bottom indicates 'Ln 143, Col 17, Ch 14, INS'.

8. Last thing before you go

Hopeless, Missouri:

```
#include "library.h"
```

```
int length_of_a_month(const int month)
```

```
{  
    if (month == 1)  
        return 31;  
    if (month == 2)  
        return 28;  
    if (month == 3)  
        return 31;  
    if (month == 4)  
        return 30;  
    if (month == 5)  
        return 31;  
    if (month == 6)  
        return 30;  
    if (month == 7)  
        return 31;  
    if (month == 8)  
        return 31;  
    if (month == 9)  
        return 30;  
    if (month == 10)  
        return 31;  
    if (month == 11)  
        return 30;  
    if (month == 12)  
        return 31;  
  
    return 0;  
}
```

```
int day_of_the_year(const int month, const int day)
```

```
{  
    if (month == 1) return day;  
    return length_of_a_month(month - 1) + day_of_the_year(month - 1, day);  
}
```

```

void Temps(ifstream & fin, double Xmax, double Ymax, double Xmin, double Ymin, double
Xwind, double Ywind, double & highest_temp, double & lowest_temp)
{
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
>> wind;

    if (fin.fail())
        return;

    if (max_temp > highest_temp)
        highest_temp = max_temp;
    if (min_temp < lowest_temp)
        lowest_temp = min_temp;

    double x = ((day_of_the_year(month,day)) * 1.8) + 50;
    double y_max = 240 - max_temp * 4;
    double y_min = 240 - min_temp * 4;
    double y_wind = 400 - wind * 4;

    set_pen_width(3);

    set_pen_color(color::red);
    move_to(Xmax, Ymax);
    draw_to(x, y_max);

    set_pen_color(color::blue);
    move_to(Xmin, Ymin);
    draw_to(x, y_min);

    set_pen_color(color::light_blue);
    move_to(Xwind, Ywind);
    draw_to(x, y_wind);

    Temps(fin, x, y_max, x, y_min, x, y_wind, highest_temp, lowest_temp);
}

void x_axis_month(int month)
{
    if (month > 12)

```

```

        return;

double x = (day_of_the_year(month, 1) * 1.8) + 50;

set_pen_color(color::grey);
move_to(x, 40.0);
draw_to(x, 440.0);

string initial;
if (month == 1)
    initial = "J";
if (month == 2)
    initial = "F";
if (month == 3)
    initial = "M";
if (month == 4)
    initial = "A";
if (month == 5)
    initial = "M";
if (month == 6)
    initial = "J";
if (month == 7)
    initial = "J";
if (month == 8)
    initial = "A";
if (month == 9)
    initial = "S";
if (month == 10)
    initial = "O";
if (month == 11)
    initial = "N";
if (month == 12)
    initial = "D";

double next_x = (day_of_the_year(month + 1, 1) * 1.8) + 50;

if (month == 12)
    next_x = (365 * 1.8) + 50;
else
    next_x = (day_of_the_year(month + 1, 1) * 1.8) + 50;

double letter = (x + next_x) / 2;

```

```

        set_pen_color(color::magenta);
        set_font_size(20);
        move_to(letter - 5, 460.0);

        write_string(initial);
        x_axis_month(month + 1);
    }

```

```

void y_axis_temp(int temp)

```

```

{
    if (temp > 100)
        return;

    double y = 440 - temp * 4;

    set_pen_color(color::black);
    move_to(50.0, y);
    draw_to(709.0, y);

    set_pen_color(color::magenta);
    set_font_size(20);
    move_to(20.0, y);
    write_string(temp);

    y_axis_temp(temp + 10);
}

```

```

void max_and_min(double highest_temp, double lowest_temp)

```

```

{
    double y_high = 240 - highest_temp * 4;
    double y_low = 240 - lowest_temp * 4;

    set_pen_width(2);
    set_pen_color(color::orange);

    move_to(50.0, y_high);
    draw_to(709.0, y_high);
    move_to(50.0, y_low);
    draw_to(709.0, y_low);
}

```

```
void graph()
```

```
{  
    make_window(800, 600);  
  
    set_pen_color(color::black);  
    move_to(50.0, 40.0);  
    draw_to(50.0, 440.0);  
    move_to(709.0, 40.0);  
    draw_to(709.0, 440.0);  
  
    x_axis_month(1);  
    y_axis_temp(0);  
}
```

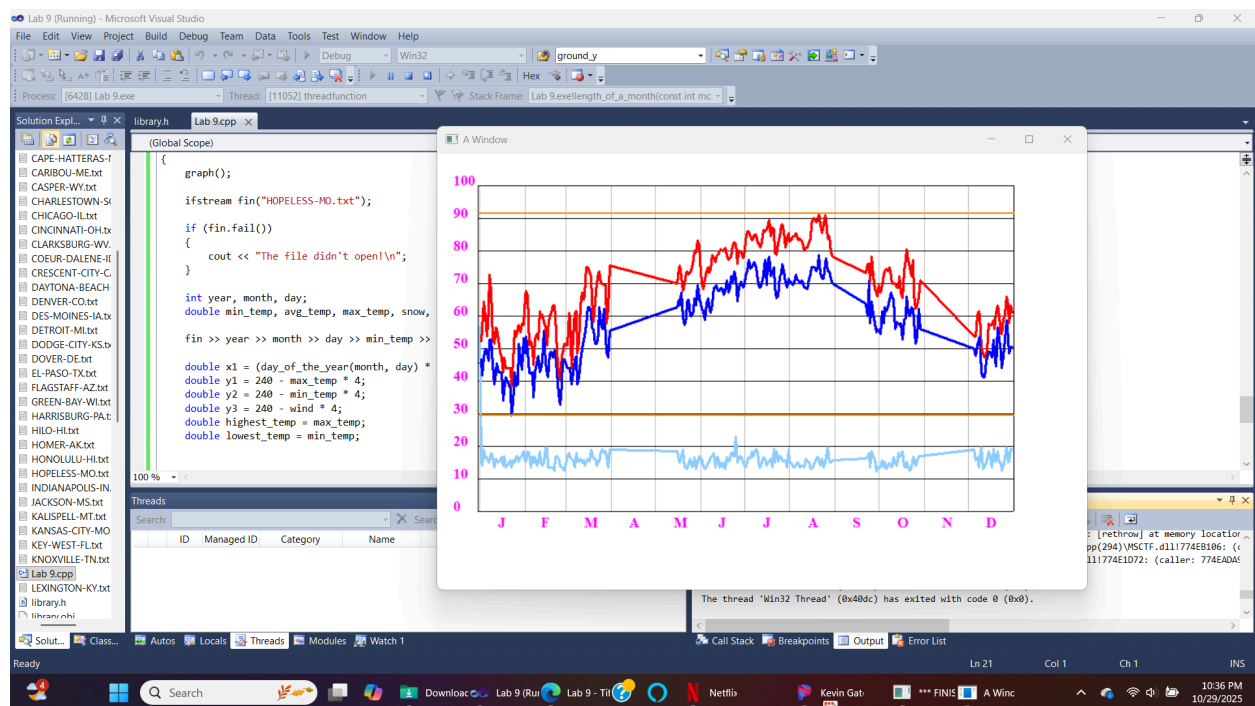
```
void main()
```

```
{  
    graph();  
  
    ifstream fin("HOPELESS-MO.txt");  
  
    if (fin.fail())  
    {  
        cout << "The file didn't open!\n";  
    }  
  
    int year, month, day;  
    double min_temp, avg_temp, max_temp, snow, rain, wind;  
  
    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain  
>> wind;  
  
    double x1 = (day_of_the_year(month, day) * 1.8) + 50;  
    double y1 = 240 - max_temp * 4;  
    double y2 = 240 - min_temp * 4;  
    double y3 = 240 - wind * 4;  
    double highest_temp = max_temp;  
    double lowest_temp = min_temp;  
  
    Temps(fin, x1, y1, x1, y2, x1, y3, highest_temp, lowest_temp);  
  
    fin.close();  
}
```

```

    max_and_min(highest_temp, lowest_temp);
}

```



Detroit, Michigan:

```
#include "library.h"
```

```

int length_of_a_month(const int month)
{
    if (month == 1)
        return 31;
    if (month == 2)
        return 28;
    if (month == 3)
        return 31;
    if (month == 4)
        return 30;
    if (month == 5)
        return 31;
    if (month == 6)
        return 30;
    if (month == 7)
        return 31;
}

```

```

        if (month == 8)
            return 31;
        if (month == 9)
            return 30;
        if (month == 10)
            return 31;
        if (month == 11)
            return 30;
        if (month == 12)
            return 31;

        return 0;
    }

```

```

int day_of_the_year(const int month, const int day)
{
    if (month == 1) return day;
    return length_of_a_month(month - 1) + day_of_the_year(month - 1, day);
}

```

```

void Temps(ifstream & fin, double Xmax, double Ymax, double Xmin, double Ymin, double
Xwind, double Ywind, double & highest_temp, double & lowest_temp)
{
    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
    >> wind;

    if (fin.fail())
        return;

    if (max_temp > highest_temp)
        highest_temp = max_temp;
    if (min_temp < lowest_temp)
        lowest_temp = min_temp;

    double x = ((day_of_the_year(month, day)) * 1.8) + 50;
    double y_max = 240 - max_temp * 4;
    double y_min = 240 - min_temp * 4;
    double y_wind = 400 - wind * 4;
}

```

```

    set_pen_width(3);

    set_pen_color(color::red);
    move_to(Xmax, Ymax);
    draw_to(x, y_max);

    set_pen_color(color::blue);
    move_to(Xmin, Ymin);
    draw_to(x, y_min);

    set_pen_color(color::light_blue);
    move_to(Xwind, Ywind);
    draw_to(x, y_wind);

    Temps(fin, x, y_max, x, y_min, x, y_wind, highest_temp, lowest_temp);
}

```

```

void x_axis_month(int month)
{
    if (month > 12)
        return;

    double x = (day_of_the_year(month, 1) * 1.8) + 50;

    set_pen_color(color::grey);
    move_to(x, 40.0);
    draw_to(x, 440.0);

    string initial;
    if (month == 1)
        initial = "J";
    if (month == 2)
        initial = "F";
    if (month == 3)
        initial = "M";
    if (month == 4)
        initial = "A";
    if (month == 5)
        initial = "M";
    if (month == 6)
        initial = "J";
    if (month == 7)
        initial = "J";
}

```

```

        if (month == 8)
            initial = "A";
        if (month == 9)
            initial = "S";
        if (month == 10)
            initial = "O";
        if (month == 11)
            initial = "N";
        if (month == 12)
            initial = "D";

        double next_x = (day_of_the_year(month + 1, 1) * 1.8) + 50;

        if (month == 12)
            next_x = (365 * 1.8) + 50;
        else
            next_x = (day_of_the_year(month + 1, 1) * 1.8) + 50;

        double letter = (x + next_x) / 2;

        set_pen_color(color::magenta);
        set_font_size(20);
        move_to(letter - 5, 460.0);

        write_string(initial);
        x_axis_month(month + 1);
    }

void y_axis_temp(int temp)
{
    if (temp > 100)
        return;

    double y = 440 - temp * 4;

    set_pen_color(color::black);
    move_to(50.0, y);
    draw_to(709.0, y);

    set_pen_color(color::magenta);
    set_font_size(20);
    move_to(20.0, y);

```

```

        write_string(temp);

        y_axis_temp(temp + 10);
    }

void max_and_min(double highest_temp, double lowest_temp)
{
    double y_high = 240 - highest_temp * 4;
    double y_low = 240 - lowest_temp * 4;

    set_pen_width(2);
    set_pen_color(color::orange);

    move_to(50.0, y_high);
    draw_to(709.0, y_high);
    move_to(50.0, y_low);
    draw_to(709.0, y_low);
}

void graph()
{
    make_window(800, 600);

    set_pen_color(color::black);
    move_to(50.0, 40.0);
    draw_to(50.0, 440.0);
    move_to(709.0, 40.0);
    draw_to(709.0, 440.0);

    x_axis_month(1);
    y_axis_temp(0);
}

void main()
{
    graph();

    ifstream fin("DETROIT-MI.txt");

    if (fin.fail())
    {

```

```

        cout << "The file didn't open!\n";
    }

    int year, month, day;
    double min_temp, avg_temp, max_temp, snow, rain, wind;

    fin >> year >> month >> day >> min_temp >> avg_temp >> max_temp >> snow >> rain
    >> wind;

    double x1 = (day_of_the_year(month, day) * 1.8) + 50;
    double y1 = 240 - max_temp * 4;
    double y2 = 240 - min_temp * 4;
    double y3 = 240 - wind * 4;
    double highest_temp = max_temp;
    double lowest_temp = min_temp;

    Temps(fin, x1, y1, x1, y2, x1, y3, highest_temp, lowest_temp);

    fin.close();

    max_and_min(highest_temp, lowest_temp);
}

```

